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Analysis of muscle architecture variables and their relationship with functional capacity in patients with fibromyalgia: A cross-sectional study

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Abstract

Background: Fibromyalgia (FM) is a chronic syndrome characterised by widespread musculoskeletal pain, accompanied by fatigue, functional alterations, and changes in body composition. **Objective:** The main objective was to analyse muscle architecture variables using ultrasound in patients with FM. **Design:** A cross-sectional observational study conducted in accordance with the STROBE guideline. **Methods:** 80 patients with FM (54.06 ± 9.66 years) were assessed. Sociodemographic and anthropometric variables, functional capacity (5-sit-to-stand-test (STST), Handgrip (HGS), Time Up Go (TUG) and 4-metres-walk-test (4MWT) and muscle architecture variables (cross-sectional-area of vastus lateralis (VL-CSA) and cross-sectional-area of rectus femoralis (RF-CSA), thickness of vastus lateralis and pennation angle of vastus lateralis (AP-VL)) were collected. Normality and variance homogeneity were assessed, and Pearson or Spearman correlations were applied as appropriate. Multiple linear regression was performed to evaluate the association between VL-CSA and clinical, functional, and anthropometric variables. Statistical significance was set at $p < 0.05$. **Results:** Participants reported high pain intensity (VAS: 7.03 ± 1.94). Mean VL-CSA was 11.73 ± 3.76 cm^2. Ultrasound-derived muscle parameters showed significant positive associations with lean mass and anthropometric variables, with a stronger association between VL-CSA and trochanteric circumference than with body mass index (BMI). The regression model explained 46.6% of VL-CSA variance ($p < 0.001$), with BMI, trochanteric circumference, 5-STST, performance and handgrip strength emerged as independent predictors ($p < 0.05$), while gait speed was not significant. **Conclusions:** Correlations were found between anthropometric variables and muscle architecture. VL-CSA was significantly and independently related to BMI, 4-meter test, 5 sit-stand test, handgrip strength, and trochanter circumference.

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Plain language summary

Fibromyalgia is a long-term condition that causes widespread pain, fatigue, and difficulties in daily activities. Many women with fibromyalgia also experience reduced muscle strength and physical function, but the role of muscle structure in these limitations is not well understood. This study included 80 women with fibromyalgia. We assessed body measurements, physical function using simple strength and mobility tests, and thigh muscle size using ultrasound, a safe and painless imaging technique. The results showed that participants reported high levels of pain and had reduced physical performance. Women with larger thigh muscles tended to have more muscle mass and better results in strength and mobility tests. In addition, body weight, hip-related measurements, and physical performance were important factors associated with muscle size. These findings suggest that muscle structure is closely related to physical function in women with fibromyalgia and that ultrasound may be a useful tool to better understand functional limitations in this population.

Keywords

fibromyalgia, ultrasound, cross-sectional-area, body composition and functional capacity

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Introduction

Fibromyalgia (FM) is a chronic syndrome characterised by widespread musculoskeletal pain that is usually accompanied by fatigue, functional alterations, sleep disorders and changes in body composition.[1]

FM is a major cause of chronic musculoskeletal pain, affecting approximately 2-4% of the general population, with a marked predominance in women.[2] It is unclear aetiology complicates both diagnosis and treatment.[3] The substantial impact of FM on daily functioning underscores its relevance as a public health concern.[4] Current management includes non-pharmacological approaches such as exercise, physiotherapy, and cognitive–behavioural therapy, often combined with pharmacological treatment.[5]

Some of the symptoms of the condition, such as fatigue and muscle pain, suggest skeletal muscle involvement. Several studies have evaluated morphological and metabolic changes in skeletal muscle through biopsy, as well as peripheral mechanisms that may play an important role in chronic pain.[6,7] This chronic pain leads patients with FM to significantly reduce their daily physical activity, which has direct consequences on body composition and functional capacity.[8]

Assessing body composition and functional capacity is essential for understanding the progression of FM, but few studies have evaluated morphological changes in skeletal muscle using ultrasound. However, ultrasound is undoubtedly a suitable tool that allows us to detect functional and structural changes in skeletal muscle.[9,10] These functional and morphological changes caused by physical inactivity led to a loss of muscle mass and strength, as well as higher pain. A higher level of body fat is known to be negatively correlated with the severity of FM.[11]

Ultrasound-derived variables quadriceps thickness and cross-sectional area (CSA) are key correlates of muscle strength.[11] Reduced CSA has been associated with greater fatigue and poorer quality of life in both healthy individuals and women with FM, with additional links to increased social isolation in the latter group.[12]

Various studies have shown that people with FM have lower functional capacity and physical performance than the general population.[13–16] Significant correlations have also been observed between the impact of FM (FIQ) and the Visual Analogue Scale (VAS) with different functional tests in women diagnosed with FM.[17]

Despite growing evidence of muscle function deterioration, reduced physical capacity, and altered body composition in women with FM, the structural characteristics of skeletal muscle and their functional relevance remain insufficiently explored. Ultrasound measurements, such as VL-CSA and thickness of vastus lateralis, offer a non-invasive and reliable method for quantifying muscle morphology; however, their association with functional performance and anthropometric indicators in this population has not been clearly established to date.

Although peripheral muscle alterations have been described in some studies, FM is primarily understood as a disorder of central sensitization rather than a primary muscle disease. Therefore, any structural or morphological differences observed in skeletal muscle should be interpreted cautiously, as they may reflect secondary consequences of chronic pain, fatigue, and reduced physical activity rather than intrinsic muscle pathology. Clarifying whether ultrasound-derived muscle characteristics relate to functional performance may help contextualize these secondary changes within the broader clinical picture of FM.

The main objective of the study was to analyse muscle architecture variables using ultrasonography in patients with FM. Secondary objectives were to correlate ultrasound-derived variables with physical capacity, assessed through functional tests

and to determine whether there is a relationship between ultrasound measures and anthropometric variables assessed through bioimpedance.

Methods

Study design and participants

The study sample consisted of 80 patients diagnosed with FM according to the 2016 American College of Rheumatology (ACR) criteria. Participants were aged between 18 and 75 years and were members of the Afinsyfacro Association in Móstoles (Madrid, Spain). All individuals who met the inclusion criteria and agreed to participate voluntarily were included. The research was conducted between 31 January 2025 and 15 December 2025. As this was a descriptive observational study, the criteria set out in the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines were followed.[18]

The inclusion criteria were as follows: (a) adults with a confirmed medical diagnosis of FM; (b) aged between 18 and 75 years; (c) sufficient cognitive ability and comprehension to complete the self-administered questionnaires.

Exclusion criteria were: (i) presence of interventions or injuries that could interfere with the ultrasound image; (ii) inability to perform functional assessments; (iii) pregnancy at the time of the study; (iv) recent surgical procedures or traumatic that could impair mobility or alter physical functioning; (v) neurological conditions affecting mobility and pain perception; (vi) metabolic disorders with a possible influence on body composition; (vii) patients with implanted pacemakers.

Questionnaires

FIQ: The Spanish version of this questionnaire was used to assess the impact of FM on functional capacity and quality of life. Scores range from 0 to 100, with 0 representing the best condition and 100 the worst (Supplementary Material 1).[19]

Hospital Anxiety and Depression Scale (HADS-A/HADS-D): we used these questionnaires to assess the symptoms of anxiety and depression. They consist of 14 items divided into two subscales of seven items each, with each item scored between 0 to 3. Total scores range from 0 to 7 (normal), 8 to 10 (mild case), and $\geq$ 11 (clinical case).[20]

Pittsburgh Sleep Quality Index (PSQI): we used this questionnaire to assess the sleep quality. This questionnaire comprises 24 items each scored from 0 to 3, where 0 indicates no difficulty and 3 indicates severe difficulty. The global score (0–21) classifies respondents as good sleepers ($\leq$5) or poor sleepers (>5) (Supplementary Material 2).[21]

Functional tests and pain

Handgrip Strength (HGS): Participants sat with their arms resting on armrests and held a digital dynamometer (Saehan DHD-1) with their elbow flexed at 90°, their wrist in 0-30° dorsiflexion, and 0-15° ulnar deviation. Three maximum efforts of 5 seconds each were performed with the dominant hand, interspersed with 30-second rest intervals. Handgrip strength values were recorded in kilograms, and the mean of the three trials was used for analysis.[22]

Chair test (5-STST): Participants began seated on a standardised chair (46 cm high) with their feet flat on the floor and their arms crossed over. They were instructed to stand up and sit down completely five times consecutively as quickly as possible without using their arms. The time, in seconds, required to complete the five repetitions was recorded from the moment the investigator gave the 'go' signal[22]

Gait speed (4-MWT): This test was used to assess walking speed (m/s). Participants were instructed to walk 4 m. as fast as possible after the 'Ready, go' signal. The time was recorded in seconds, and the mean of the two attempts was used for analysis.[22]

Timed Up Go (TUG): Participants were instructed to stand up from a 46 cm-high chair, walk 3 m, turn around a cone, return to the chair, and sit down. Two attempts were performed, with a 3-minute rest period between trials. The test began following the 'go' signal, and the time taken to complete the task was recorded by the investigators. The time taken to complete the test was recorded in seconds, and the mean of the two attempts was calculated for analysis.[23]

Algometry: to assess pain sensitivity, an analogue algometer (Fischer FPK 20) applied to specific tender points (bilateral epicondyle, inner knee, and trochanter). Three measurements were taken at $\geq$30-s intervals, and the mean value was calculated.[24] Pain intensity was evaluated using a VAS, a 10 cm line on which participants indicated their perceived pain intensity using 0-10 scale.[25]

Anthropometric and ultrasound measurements

Body composition analysis: Weight and body composition were assessed using direct segmental multi-frequency bioelectrical impedance analysis (InBody® 770, Bilbao, Spain), applying six frequencies (1, 5, 50, 250, 500, and 1000 kHz). This technique provides accurate estimates of fat mass, lean mass, and body water content and distribution.[26]

Muscle architecture measurements: Participants were positioned in the supine position with the limbs relaxed and aligned in the anatomical position before ultrasound assessment. The site used to evaluate vastus lateralis (VL) muscle architecture was determined at the midpoint (50%) between the greater trochanter and the knee joint line. The ultrasound probe was positioned with a neutral orientation, perpendicular to the longitudinal axis of the limb. A layer of conductive gel was applied to the skin before the probe made contact to facilitate image acquisition. For each variable, three ultrasound images were obtained, and the average of these measurements was used for subsequent analysis. The pennation angle (AP) was calculated as the angle between the muscle fascicle and the deep aponeurosis. The CSA was assessed in panoramic images by freehand planimetry of the VL belly within the aponeuroses (Figure 1).[11,27] Muscle thickness was defined as the linear distance between the superficial and deep aponeuroses of each muscle evaluated. Measurements were obtained by drawing a perpendicular line between these two structures at the point of greatest thickness observed in the ultrasound image 30 Three independent images were acquired at each anatomical site, and the mean value of these three measurements was used for subsequent statistical analysis. All ultrasound scans were performed by a single investigator (E.C-F-P) with more than 10 years of prior experience in musculoskeletal ultrasound. Image acquisition and data extraction were performed by another independent investigator. Neither had access at any time to the participants' clinical or functional data. Although reliability was not directly assessed in the present study, all measurements were conducted under strictly standardised conditions by experienced operators, following previously described protocols that have demonstrated high reliability.[28–30]

Ethical issues

The study protocol was reviewed and approved by the Ethics Committee of Hospital Clínico San Carlos (approval number 24/745- EC_X) and was registered on https://clinicaltrials.gov/ under the identifier NCT06253273.

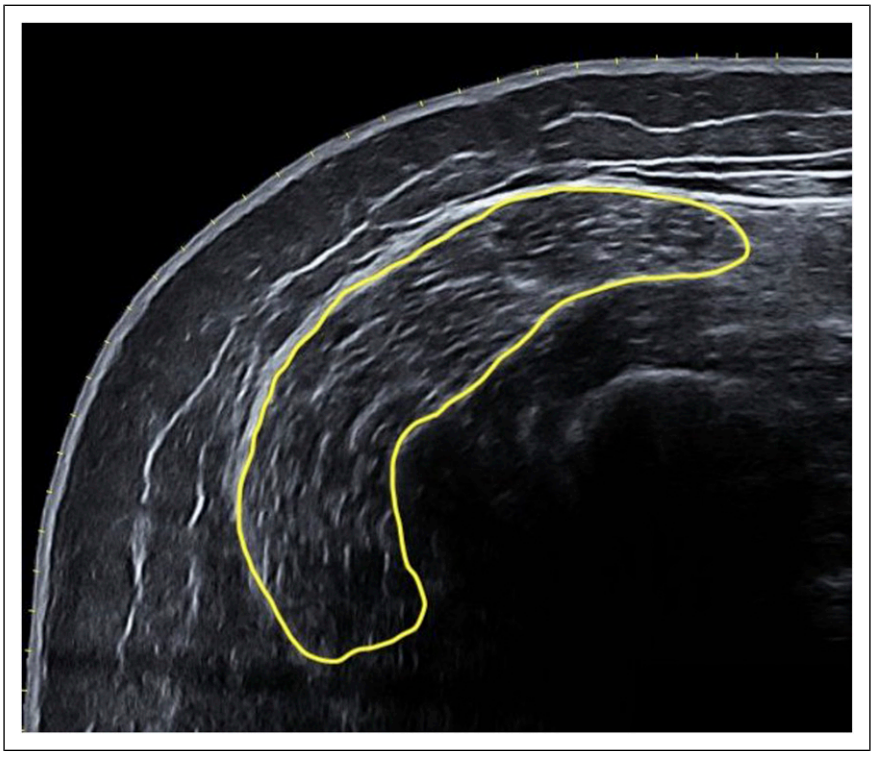


Figure 1. Ultrasound image of VL-CSA.

All procedures adhered to the principles outlined in the 2014 Declaration of Helsinki of the World Medical Association concerning medical research involving human subjects. Participants' privacy and confidentiality were rigorously protected, and written informed consent was obtained from all participants; documentation is securely held by the corresponding author.

Statistical analysis

Data were analysed using SPSS software, version 29.0. Categorical variables are presented as frequencies and percentages, while continuous variables are reported as means ± standard deviation (SD). The normality of continuous variables was assessed using the Kolmogorov–Smirnov test. Homogeneity of variances was evaluated prior to performing parametric tests. Associations between variables were examined using Spearman's rho for non-parametric data and Pearson's correlation coefficient for parametric data. Where appropriate, multiple comparisons were adjusted to control for Type I error. Statistical significance was set at $p < 0.05$. Outliers were identified and examined for plausibility; extreme values were retained only if they reflected valid measurements.

All analyses were conducted according to assumptions of the selected statistical tests, ensuring robustness and reproducibility of the results. A multivariate lineal regression model was used to analyze the relationship between the variable of interest, VL-CSA and the other variables in the study, including all participants, regardless of their sex. A multicollinearity analysis was performed using the VIF. The VIF values below 5 for the model indicate the absence of significant multicollinearity. The multivariate lineal model was calculated using the "stepwise forward" strategy.

Results

Participants had a mean age of 54.06 ± 9.66 years, with a predominance of women (96.3%). The mean pain intensity reported using the VAS was 7.03 ± 1.94. Sociodemographic characteristics of the study sample are summarised in Table 1.

Descriptive results for functional performance, pain sensitivity, limb muscle mass, and ultrasound-derived variables are presented in Table 2. The table also includes mean architectural measurements, VL-CSA: 11.73 ± 3.76 cm^2, RF-CSA: 6.19 ± 2.20 cm^2 and AP-VL 13.70 ± 3.04°.

Spearman's correlation analysis showed positive and statistically significant associations between ultrasound-derived muscle parameters and markers of lean mass and anthropometry (Table 3).

Figure 2 illustrates the bivariate associations with VL-CSA. A positive and more consistent linear trend was observed with trochanteric circumference, whereas the association with BMI was also positive but more dispersed. Overall, the pattern indicates a stronger relationship between VL-CSA and trochanteric circumference than with BMI.

A multiple linear regression model was conducted with VL-CSA as the dependent variable. The final model explained a substantial proportion of the variance (adjusted $R^2 = 0.466$; standard error = 2.90), indicating that approximately half of the variability in VL-CSA was accounted for by the included predictors. The overall model fit was statistically significant ($F = 9.44$, $p < 0.001$). BMI, trochanteric circumference, marital status, 5-STST and handgrip strength emerged as independent predictors of VL-CSA (all $p < 0.05$). Among these, BMI showed the strongest standardized association ($\beta = 0.344$, $p = 0.012$). Gait speed in the 4-MWT did not reach statistical significance ($\beta = 0.181$, $p = 0.064$) (Table 4).

Table 1. Sociodemographic and descriptive variables of the participants.

Variable	Mean ± SD/%
Age	54.06 ± 9.66
Marital status (a-d)	56.3%;22.5%;20.0%;1.3%
Menstrual status (e-h)	22.5%;20.0%;53.8%;3.8%
Smoking (i,j)	20.2%;78.8%
Alcohol (k-m)	1.3%;36.3%;61.3%
VAS (0-10)	7.03 ± 1.94
HADS-Anxiety (0-21)	12.65 ± 4.56
HADS-Depression (0-21)	12.65 ± 4.56

VAS: Visual Analogue Scale; HADS-A: Hospital Anxiety Depression Scale; a: married; b: separated; c: single; d: widowed; e: menstruating; f: I am a woman who does not menstruate; g: menopausal; h: men; i: smoker; j: no smoker; k: yes, daily l: yes, sometimes (1 or 2 times a week); m: no alcohol.

Table 2. Descriptive analysis of functionality, pain sensitivity, body composition and muscular architectural measures.

Variable	Mean ± SD
HGS	21.17 ± 8.49
HGS (female)	20.21 ± 6.35
HGS (men)	46.59 ± 18.19
5-STST (s)	12.83 ± 5.70
4-MWT (m/s)	1.01 ± 0.56
TUG (s)	9.72 ± 7.10
FIQ (0-100)	65.27 ± 16.07
PSQI (0-21)	14.88 ± 4.29
Algometry	
Left epicondyle	1.70 ± 0.70
Right epicondyle	1.70 ± 0.75
Left trochanter	1.99 ± 0.93
Right trochanter	1.90 ± 0.94
Left knee	1.86 ± 0.92
Right knee	1.88 ± 0.86
BMI (kg/m2)	27.47 ± 5.66
TBW (kg)	32.56 ± 4.89
Protein (kg)	8.64 ± 1.31
FMB (kg)	27.49 ± 11.32
FFM (kg)	44.37 ± 6.65
SMM (kg)	24.08 ± 3.96
SMM female (kg)	23.75 ± 3.37
SMM men (kg)	34.06 ± 2.87
PBF (%)	37.02 ± 8.34
WHR	0.95 ± 0.07
VFL	13.02 ± 5.07
Hip circumference (cm)	105.69 ± 12.38
Trochanter circumference (cm)	53.38 ± 7.31
Waist circumference (cm)	92.15 ± 13.45
Mean VL-CSA	11.73 ± 3.76
Mean VL-CSA (female)	11.71 ± 3.75
Mean VL-CSA (men)	13.46 ± 4.35
Mean RF CSA	6.19 ± 2.20
Mean AP VL	13.70 ± 3.04
Mean AP VL (female)	13.58 ± 3.01
Mean AP VL (men)	16.23 ± 3.83
Mean thickness VL (female)	1.63 ± 0.28
Mean thickness VL (men)	2.16 ± 0.37
Thickness VL	1.65 ± 0.30

HGS: Handgrip Strength; 5-STST: 5-sit-to-stand-test; 4-MWT: 4-metres walk-test; TUG: Time Up Go; FIQ: Fibromyalgia Impact Questionnaire; PSQI: Pittsburgh Sleep Quality Index; BMI: Body Mass Index; TBW: Total Body Water; SMM: skeletal muscle mass; PBF: percentage body fat; FBM: Fat Body Mass; FFM: Fat Free Mass; ant leg; WHR: Waist-Hip Ratio; VFL: Visceral Fat Level; CSA: cross-sectional area VL: vastus lateralis; RF: rectus femoris; AP: angle of pennation.

Discussion

The main objective of this study was to analyse muscle architecture variables (specifically VL-CSA, RF-CSA and VL thickness) in women with FM, and to examine their associations with anthropometric characteristics and functional capacity. Overall, the findings show moderate associations between muscle architecture variables and functional tests (5-STST and HGS), suggesting that these measures are related to physical performance in women with FM.

Table 3. Correlations between ultrasound variables and body composition variables.

	FFMDL	WDL	Protein	SMM	TrochCir
VL-CSA	0.410*	0.419*	0.408*	0.404*	0.499*
RF-CSA	0.468*	0.469*	0.449*	0.432*	0.420*

CSA: cross-sectional area VL: vastus lateralis; RF: rectus femoris; FFMDL: FFM of dominant leg; WDL: water of dominant leg; SMM: Muscle mass of the limbs; TrochCir: trochanter circumference *p < 0.01.

The functional profile observed, characterised by impaired lower-limb performance alongside relatively preserved handgrip strength, suggests a differential impact of FM on weight-bearing versus non-weight-bearing tasks. Specifically, performance in the 5-STST and TUG indicated reduced physical capacity,[31,32] whereas handgrip strength values were approximately at the 50th percentile for age- and sex-matched populations.[33] Gait speed was also below normative values for women aged 50–59 years.[34] In addition, the sample presented an overweight profile with high fat percentage and relatively low skeletal muscle mass (SMM), consistent with previous reports in Spanish cohorts of women with FM.[35]

Although reference values for muscle architecture variables in populations with FM remain scarce, parameters such as VL-CSA, muscle thickness, and AP have been characterised in other cohorts. In healthy women aged 50–60 years, VL thickness of 1.9 ± 0.5 cm and 1.89 ± 0.27 cm have been described, while AP of 17.3 ± 4.9° and 12.19 ± 1.46°, respectively, have been described.[36,37] Regarding CSA, values reported in healthy women of similar age are lower than those observed in

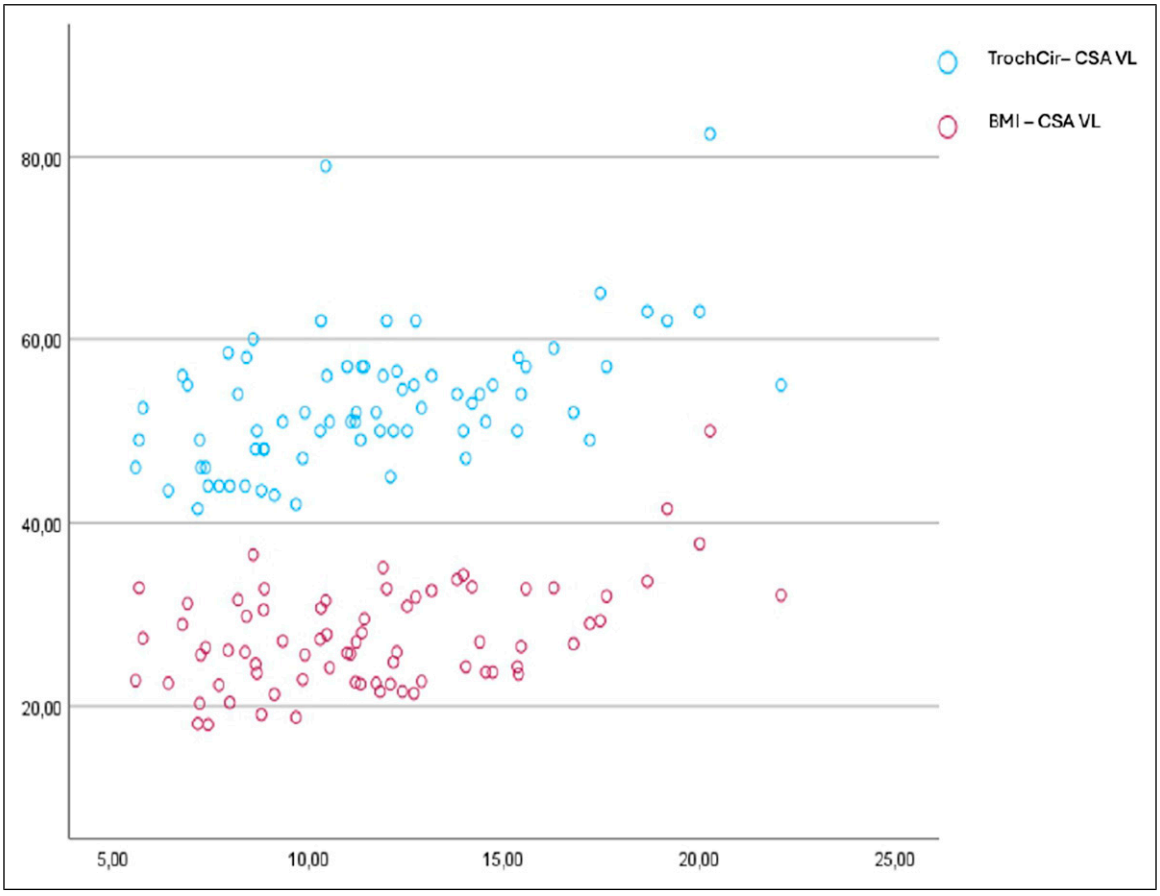


Figure 2. Correlation between BMI, TrochCir and mean of VL-CSA. BMI: Body Mass Index; VL- CSA: cross-sectional area of vastus laterallis.

Table 4. Multivariable lineal regression model.

Variable	Unstandardized coefficient beta	Standardized beta coefficients	*p*-value	Confidence interval
BMI	0.228	0.344	0.012	0.40 – 0.005
TrochCir	0.137	0.266	0.046	0.27 – 0.002
Marital Status	1.250	0.270	0.004	2.09 – 0.41
5-STST	0.140	0.226	0.020	0.27 – 0.02
HGS	0.100	0.230	0.016	0.18 – 0.02
4-MWT	1.190	0.180	0.064	2.45 – (-0.07)

BMI: Body Mass Index; 4-MWT:4-metres-test-walk;5-STST: 5-sit-to-stand-test;HGS: Handgrip Strenght; TrochCir: trochanter circumference.

our sample, suggesting that RF-CSA in this cohort is not reduced despite functional impairment.[36] In patellofemoral pain syndrome, significantly lower VL-CSA values have been observed compared with healthy controls.[38]

It has been demonstrated that VL-CSA and thickness of VL are directly related to muscle strength.[39,40] Previous studies have reported associations between VL-CSA, VL thickness and muscle strength measures in different population, in women with osteoarthritis, lower knee extension strength has been associated with a greater risk of worsening knee pain. This population reports greater pain and inability to perform knee extension.[41] In our sample, relationships between the HGS and 5-STST functional tests have been observed, even though these patients do not have this condition.

In our sample, VL-CSA showed statistically significant correlations with 5-STST and HGS. The same occurs in older populations, where a correlation is observed between VL-CSA with stride length and gait speed.[42]

On the other hand, and in relation to body composition parameters, we found that the VL-CSA and RF-CSA correlates significantly ($p<0.001$) with the anthropometric variables FFMDL WDL, SMM and trochanter circumference. A study in obese population showed that RF-CSA is a strong predictor of active cell mass, along with HGS and BMI.[43] Consistently, in our regression model, VL-CSA was significantly associated with both anthropometric and functional factors. BMI emerged as the strongest predictor, suggesting that greater VL-CSA is associated with higher body weight.[44,45] Importantly, a larger CSA does not necessarily imply better muscle quality, particularly in patients with FM, because CSA reflects structural size rather than contractile function.[46,47] The significant association with trochanter circumference supports the use of ultrasound as a valid tool for assessing muscle morphology and its potential role as a structural biomarker in FM.[48]

The observed associations between VL-CSA and functional tests (5-STST and HGS) suggest that muscle architecture variables and functional performance may be related in women with FM. However, the cross-sectional nature of the study precludes determining whether these morphological characteristics influence functional capacity or simply coexist with other factors associated with physical condition.[49] Previous studies have found correlations between ultrasound-derived measures and absolute strength in the upper and lower limbs. In healthy women, we found a correlation between leg extension strength and VL-CSA, but no significant correlation was found with HGS.[50] Interestingly, no significant association was observed between pain intensity and muscle architecture variables. This finding suggests that pain and physical capacity may represent partially independent dimensions in FM, supporting previous evidence that functional impairment is not solely driven by pain severity[51]

Importantly, the cross-sectional design does not allow causal inferences regarding the relationship between muscle architecture and functional performance. Therefore, the observed associations should not be interpreted as evidence that variations in muscle morphology determine physical capacity, as both may be influenced by other external factors.

From a clinical perspective, these findings support the use of ultrasound as a complementary tool for assessing muscle morphology and physical condition in women with FM.[52] Ultrasound-derived variables, such as VL-CSA, VL thickness and AP VL, provide objective information on muscle mass and architecture, which could help clinicians identify patients with alterations in muscle mass and low physical capacity. Given the observed associations between VL-CSA and functional outcomes, particularly handgrip strength and the 5-STST, ultrasound assessment may contribute to the early detection of muscle deterioration and the monitoring of adaptations to exercise-based interventions. As ultrasound is a non-invasive, low-cost and widely accessible imaging technique, it could be used as a tool to complement patient-reported outcomes and functional performance tests in clinical practice. However, muscle morphology should not be considered a primary diagnostic target; rather, the assessment of muscle architecture can contribute to a more comprehensive evaluation of physical function.

Limitations and future lines of research

Several limitations should be acknowledged when interpreting the findings of this study. First, the sample was small and consisted predominantly of women and belonged to a single association. Second, the analysis did not incorporate information regarding participants' habitual physical activity levels or medication used or duration of FM, all these factors can affect muscle morphology and functional capacity Third, the absence of a control group prevents direct comparison with healthy individuals, limiting the ability to determine whether the observed findings are specific to FM or reflect more general patterns. Additionally, ultrasound assessments were restricted to the lower extremities, preventing a broader understanding of potential architectural alterations in upper-limb musculature. In addition, intra-rater and inter-rater reliability were not formally evaluated for the ultrasound measurements. The use of a stepwise variable-selection procedure should be considered a limitation of the present study. Although this approach was useful for exploratory model building and for identifying variables independently associated with VL-CSA, stepwise methods are data-driven and may produce sample-dependent models, unstable coefficient estimates, and an increased risk of overfitting. Furthermore, clinically relevant variables may be excluded when they do not meet the statistical criteria for entry or retention. Consequently, the associations identified in the present model should be interpreted cautiously and require confirmation using prespecified regression models and validation in independent samples. Although image acquisition and analysis were performed by experienced investigators following standardised protocols with previously reported high reliability, intra- and inter-rater reliability were not formally assessed in the present study. Consequently, measurement error and observer-related variability cannot be completely excluded, which may have influenced the precision of muscle architecture measurements and, therefore, the strength of the observed associations.

Furthermore, although muscle architecture values were interpreted in relation to previously reported reference values, the lack of well-established normative data for FM populations together with the absence of complementary indicators of muscle quality, such as echo intensity or neuromuscular activation, limits the interpretation of cross-sectional area and muscle thickness as surrogate markers of muscle function. Therefore, the apparent preservation of muscle morphology observed in this study should be interpreted with caution, as these structural parameters may be influenced by body composition, chronic pain-related adaptations, or other systemic factors rather than reflecting true muscle function.

Future research should incorporate control groups, longitudinal designs, and interventional approaches to clarify causal relationships and determine whether targeted exercise interventions can modify muscle architecture and improve functional outcomes. Expanding ultrasound assessment to additional muscle groups and integrating objective measures of physical activity would further enhance understanding of musculoskeletal adaptations in FM.

Conclusions

In this cohort of women with fibromyalgia, ultrasound-derived measures of the vastus lateralis and rectus femoris demonstrated moderate associations with anthropometric characteristics and functional performance. These relationships should be interpreted within the broader context of fibromyalgia as a condition primarily driven by central sensitization rather than intrinsic muscle pathology. The morphological differences observed in muscle architecture are therefore more likely to reflect secondary adaptations related to physical deconditioning, reduced physical activity, and chronic symptom burden.

Although ultrasound cannot determine the underlying mechanisms of these changes, it offers a non-invasive and accessible method to characterize physical status in this population. The associations identified between muscle morphology, handgrip strength, and lower-limb functional tests highlight the potential value of integrating ultrasound assessments into comprehensive evaluations of physical function. Such information may help clinicians identify individuals who could benefit from tailored exercise-based interventions aimed at improving strength, mobility, and overall conditioning.

However, the clinical interpretation of ultrasound findings must remain cautious. Given the cross-sectional design of the study, no causal inferences can be drawn, and muscle morphology should not be considered a primary therapeutic target in fibromyalgia. Instead, these findings underscore the importance of addressing physical deconditioning as part of a broader, multidisciplinary approach that acknowledges the central sensitization mechanisms underlying the syndrome. Future longitudinal studies are needed to clarify how changes in muscle architecture evolve over time and whether they respond to targeted rehabilitation strategies.

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Ethical considerations

The study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committee of Hospital Clínico San Carlos, Madrid, Spain (approval number: 24/745-EC_X, data: 5 December 2024).

Consent to participate

Written informed consent was obtained from all subjects involved in the study.

Author contributions

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The data supporting the study's findings are available from the corresponding author upon reasonable request.

Supplemental material

Supplemental material for this article is available online.

References

1. Richard JY, Hurley RA and Taber KH. *Fibromyalgia: Centralized Pain Processing and Neuroimaging*. American Psychiatric Association Publishing, 2019.
2. Sumpton JE and Moulin DE. Fibromyalgia. In: *Handbook of Clinical Neurology*. Elsevier B.V., 2014, pp. 513–527.
3. Al SS, Varga SJ, Al-Husinat L, et al. Unraveling the Complex Web of Fibromyalgia: A Narrative Review. *Medicina (Lithuania)*; 60. https://doi.org/10.3390/medicina60020272, Epub ahead of print 1 February 2024.
4. Villanueva VL, Valía JC, Cerdá G, et al. Fibromialgia: diagnóstico y tratamiento. El estado de la cuestión. *Revista Sociedad Española del Dolor*; 11: 430–443.
5. Ram PR, Jeyaraman M, Jeyaraman N, et al. Beyond the Pain: A Systematic Narrative Review of the Latest Advancements in Fibromyalgia Treatment. *Cureus* 2023. https://doi.org/10.7759/cureus.48032, Epub ahead of print 31.
6. Le Goff P. Is fibromyalgia a muscle disorder? *Joint Bone Spine* 2006; 73: 239–242. https://doi.org/10.1016/j.jbspin.2005.03.022
7. Ruggiero L, Manganelli F and Santoro L. Muscle pain syndromes and fibromyalgia: The role of muscle biopsy. *Curr Opin Support Palliat Care* 2018; 12: 382–387. https://doi.org/10.1097/SPC.0000000000000355
8. Akdoğan MR, Doğanci EB, Tüzün Z, et al. Does fibromyalgia affect physical activity due to pain? *Agri* 2023; 35: 63–67.
9. Wallwork TL, Stanton WR, Freke M, et al. The effect of chronic low back pain on size and contraction of the lumbar multifidus muscle. *Man Ther* 2009; 14: 496–500. https://doi.org/10.1016/j.math.2008.09.006
10. Mesci N and Mesci E. Association of ultrasonographic measurements of upper arm and thigh muscle thickness with functional capacity and balance in elderly individuals. *Turk Geriatri Derg* 2018; 21: 215–224.
11. Ruas CV, Pinto RS, Lima CD, et al. Test-Retest Reliability of Muscle Thickness, Echo-Intensity and Cross Sectional Area of Quadriceps and Hamstrings Muscle Groups Using B-mode Ultrasound. *International Journal of Kinesiology and Sports Science* 2017; 5: 35. https://doi.org/10.7575/aiac.ijkss.v.5n.1p.35
12. Umay E, Gundogdu I and Ozturk EA. What happens to muscles in fibromyalgia syndrome. *Ir J Med Sci* 2020; 189: 749–756. https://doi.org/10.1007/s11845-019-02138-w

13. Maquet D, Croisier J-L, Renard C, et al. *Muscle performance in patients with fibromyalgia*, 2002.
14. Jones J, Rutledge DN, Jones KD, et al. Self-Assessed Physical Function Levels Of Women with Fibromyalgia. A National Survey. *Women's Health Issues* 2008; 18: 406–412. https://doi.org/10.1016/j.whi.2008.04.005
15. Tüzün EH, Albayrak G, Eker L, et al. A comparison study of quality of life in women with fibromyalgia and myofascial pain syndrome. *Disabil Rehabil* 2004; 26: 198–202. https://doi.org/10.1080/09638280310001639696
16. Panton LB, Kingsley JD, Toole T, et al. A comparison of physical functional performance and strength in women with fibromyalgia, age- and weight-matched controls, and older women who are healthy. *Phys Ther* 2006; 86: 1479–1488. https://doi.org/10.2522/ptj.20050320
17. Úbeda-D'Ocasar E, Mayoral-Gonzalo N, Pedauyé-Rueda B, et al. Relationship between pain, functionality, and body composition in patients with fibromyalgia: cross-sectional study. *Front Med (Lausanne)*, 12: 1–8. https://doi.org/10.3389/fmed.2025.1745951, Epub ahead of print 12 January 2026.
18. Von Elm E, Altman DG, Egger M, et al. Declaración de la Iniciativa STROBE (Strengthening the Reporting of Observational studies in Epidemiology): directrices para la comunicación de estudios observacionales. https://www.epidem.com/ (2008).
19. Monterde S, Salvat I and Fernández-Ballart J. *Validación de la versión española del Fibromyalgia Impact Questionnaire*. Rev Esp Reumatol.
20. Lloyd M, Sugden N, Thomas M, et al. The structure of the Hospital Anxiety and Depression Scale: Theoretical and methodological considerations. *British Journal of Psychology* 2023; 114: 457–475. https://doi.org/10.1111/bjop.12637
21. Larche CL, Plante I, Roy M, et al. The Pittsburgh Sleep Quality Index: Reliability, Factor Structure, and Related Clinical Factors among Children, Adolescents, and Young Adults with Chronic Pain. *Sleep Disord* 2021; 2021: 1–8. https://doi.org/10.1155/2021/5546484
22. Beaudart C, Rolland Y, Cruz-Jentoft AJ, et al. Assessment of Muscle Function and Physical Performance in Daily Clinical Practice: A position paper endorsed by the European Society for Clinical and Economic Aspects of Osteoporosis, Osteoarthritis and Musculoskeletal Diseases (ESCEO). *Calcified Tissue International*, 105: 1–14. https://doi.org/10.1007/s00223-019-00545-w, Epub ahead of print 15 July 2019.
23. Podsiadlo JD, Bscpt S and Richardson MDJ. *The Timed 'Up & Go': A Test of Basic Functional Mobilitv for Frail Elderlv Persons*, 1991.
24. Úbeda-D'ocasar E, Valera-Calero JA, Hervás-Pérez JP, et al. Pain intensity and sensory perception of tender points in female patients with fibromyalgia: A pilot study. *Int J Environ Res Public Health* 2021; 18: 1–7. https://doi.org/10.3390/ijerph18041461
25. Pasqual Marques A, Assumpção P, Matsutani LA, et al. Pain in Fibromyalgia and discriminative power of the instruments: visual analog scale, dolorimetry and the mcgill pain questionnaire. *Acta Reumatol Port* 2008; 33: 345–351.
26. Looney DP, Schafer EA, Chapman CL, et al. Reliability, biological variability, and accuracy of multi-frequency bioelectrical impedance analysis for measuring body composition components. *Front Nutr* 2024; 11: 1–8. https://doi.org/10.3389/fnut.2024.1491931, Epub ahead of print.
27. Nagae M, Umegaki H, Yoshiko A, et al. Muscle ultrasound and its application to point-of-care ultrasonography: a narrative review. *Ann Med* 2023; 55: 190–197. https://doi.org/10.1080/07853890.2022.2157871
28. Ritsche P, Schmid R, Franchi MV, et al. Agreement and reliability of lower limb muscle architecture measurements using a portable ultrasound device. *Front Physiol*, 13: 1–9. https://doi.org/10.3389/fphys.2022.981862, Epub ahead of print 2 September 2022.
29. Hagoort I, Hortobágyi T, Vuillerme N, et al. Age- and muscle-specific reliability of muscle architecture measurements assessed by two-dimensional panoramic ultrasound. *Biomed Eng Online*, 21: 1–19. https://doi.org/10.1186/s12938-021-00967-4, Epub ahead of print. 1 December 2022.
30. Mechelli F, Arendt-Nielsen L, Stokes M, et al. Validity of ultrasound imaging versus magnetic resonance imaging for measuring anterior thigh muscle, subcutaneous fat, and fascia thickness. *Methods Protoc* 2019; 2: 1–10. https://doi.org/10.3390/mps2030058
31. Klukowska AM, Staartjes VE, Vandertop WP, et al. Five-Repetition Sit-to-Stand Test Performance in Healthy Individuals: Reference Values and Predictors From 2 Prospective Cohorts. *Neurospine* 2021; 18: 760–769. https://doi.org/10.14245/ns.2142750.375
32. Pereiro AX, Campos-Magdaleno M, Navarro-Pardo E, et al. Normative scores for the Timed Up & Go in a Spanish sample of community-dweller adults with preserved functionality. *Aten Primaria*, 53: 1–6. https://doi.org/10.1016/j.aprim.2021.102065, Epub ahead of print 1 August 2021.
33. Tomkinson GR, Lang JJ, Rubín L, et al. International norms for adult handgrip strength: A systematic review of data on 2.4 million adults aged 20 to 100+ years from 69 countries and regions. *Journal of Sport and Health Science*, 14: 1–17. https://doi.org/10.1016/j.jshs.2024.101014, Epub ahead of print 1 December 2025.
34. Bohannon RW and Wang YC. Four-Meter Gait Speed: Normative Values and Reliability Determined for Adults Participating in the NIH Toolbox Study. *Arch Phys Med Rehabil* 2019; 100: 509–513. https://doi.org/10.1016/j.apmr.2018.06.031
35. Aparicio VA, Ortega FB, Heredia JM, et al. Análisis de la composición corporal en mujeres con fibromialgia. *Reumatol Clin* 2011; 7: 7–12.

36. El-Ansary D, Marshall CJ, Farragher J, et al. Architectural anatomy of the quadriceps and the relationship with muscle strength: An observational study utilising real-time ultrasound in healthy adults. *J Anat* 2021; 239: 847–855. https://doi.org/10.1111/joa.13497
37. Jacob I, Johnson MI, Jones G, et al. Age-related differences of vastus lateralis muscle morphology, contractile properties, upper body grip strength and lower extremity functional capability in healthy adults aged 18 to 70 years. *BMC Geriatr*, 22: 1–16. https://doi.org/10.1186/s12877-022-03183-4, Epub ahead of print 1 December 2022.
38. Dong C, Li M, Hao K, et al. Dose atrophy of vastus medialis obliquus and vastus lateralis exist in patients with patellofemoral pain syndrome. *J Orthop Surg Res*, 16: 1–6. https://doi.org/10.1186/s13018-021-02251-6, Epub ahead of print 1 December 2021.
39. Suchomel TJ and Stone MH. The relationships between hip and knee extensor cross-sectional area, strength, power, and potentiation characteristics. *Sports*, 5: 1–9. https://doi.org/10.3390/sports5030066, Epub ahead of print 1 September 2017.
40. Şekir U, Can Yalaki U and Akova B. Rectus femoris muscle thickness and cross-sectional area on ultrasonography may predict isometric and isokinetic knee extension strength: A cross-sectional study. *Turkish Journal of Sports Medicine* 2021; 57: 21–30. https://doi.org/10.47447/tjsm.0585
41. Glass NA, Torner JC, Frey LLA, et al. The relationship between quadriceps muscle weakness and worsening of knee pain in the MOST cohort: A 5-year longitudinal study. *Osteoarthritis Cartilage* 2013; 21: 1154–1159. https://doi.org/10.1016/j.joca.2013.05.016
42. Guadagnin EC, Priario LAA, Carpes FP, et al. Correlation between lower limb isometric strength and muscle structure with normal and challenged gait performance in older adults. *Gait Posture* 2019; 73: 101–107. https://doi.org/10.1016/j.gaitpost.2019.07.131
43. García Olivares M, Dalla Rovere L, Fernández Jiménez R, et al. Body Composition and Function in Patients with Obesity in Clinical Practice: Beyond the Body Mass Index. *Obes Facts* 2025: 12–21. https://doi.org/10.1159/000546820, Epub ahead of print.
44. Liu PT, Wei TS and Ching CTS. Validation of Ultrasound Measurement of Vastus Lateralis for Appendicular Skeletal Muscle Mass in Chronic Kidney Disease Patients with Hemodialysis. *Diagnostics*, 14: 1–12. https://doi.org/10.3390/diagnostics14222600, Epub ahead of print 1 November 2024.
45. Eggelbusch M, Weide G, Tieland M, et al. Estimating vastus lateralis muscle volume from a single ultrasound image. *Sci Rep*, 15: 1–13. https://doi.org/10.1038/s41598-025-11437-5, Epub ahead of print 1 December 2025.
46. Christidis N, Ghafouri B, Larsson A, et al. Comparison of the levels of pro-inflammatory cytokines released in the vastus lateralis muscle of patients with fibromyalgia and healthy controls during contractions of the quadriceps muscle - A microdialysis study. *PLoS One*, 10: 1–17. https://doi.org/10.1371/journal.pone.0143856, Epub ahead of print. 1 December 2015.
47. Häkkinen A, Häkkinen K, Hannonen P, et al. Strength training induced adaptations in neuromuscular function of premenopausal women with fibromyalgia: comparison with healthy women.2001. https://www.annrheumdis.com
48. Salaffi F, Carotti M, Di Matteo A, et al. Ultrasound and magnetic resonance imaging as diagnostic tools for sarcopenia in immune-mediated rheumatic diseases (IMRDs). *Radiologia Medica* 2022; 127: 1277–1291. https://doi.org/10.1007/s11547-022-01560-y
49. Liu T, Xie H, Yan S, et al. Thigh muscle features in female patients with severe knee osteoarthritis: a cross-sectional study. *BMC Musculoskelet Disord*, 26: 1–13. https://doi.org/10.1186/s12891-025-08361-0, Epub ahead of print. 1 December 2025.
50. Nabavizadeh O and Herda AA. Sex-Related Differences of Weight Bearing and Non-Weight Bearing Muscle Properties. *Muscles* 2023; 2: 400–412. https://doi.org/10.3390/muscles2040031
51. Merriwether EN, Frey-Law LA, Rakel BA, et al. Physical activity is related to function and fatigue but not pain in women with fibromyalgia: Baseline analyses from the Fibromyalgia Activity Study with TENS (FAST). *Arthritis Res Ther*, 20: 1–13. https://doi.org/10.1186/s13075-018-1671-3, Epub ahead of print 29 August 2018.
52. Mesci E, Dogruoz B, Mesci N, et al. Ultrasonographic evaluation of muscle thickness in female patients with fibromyalgia and its relationship with clinical parameters. *North Clin Istanb* 2023; 10: 718–725.